Machine Learning (ML) Overview

1. Definition:

Machine Learning is a subset of Artificial Intelligence (AI) that allows systems to learn from data and improve performance without being explicitly programmed.

2. Types of ML:

- Supervised Learning: Learning from labeled data. Example: Predicting house prices.
- Unsupervised Learning: Finding patterns in unlabeled data. Example: Customer segmentation.
- Reinforcement Learning: Learning by trial and error using rewards. Example: Game playing AI.

3. Popular Algorithms:

- Linear Regression
- Decision Trees
- Random Forest
- Support Vector Machines (SVM)
- K-Means Clustering
- Neural Networks

4. Applications:

- Image and Speech Recognition
- Natural Language Processing (NLP)
- Recommender Systems
- Autonomous Vehicles

5. Benefits:

- Automates decision-making
- Handles large amounts of data
- Improves accuracy over time

6. Challenges:

- Requires large datasets

- Risk of bias
- Interpretability issues

Diagram: Example of Supervised Learning Workflow

[Data Collection -> Data Preprocessing -> Model Training -> Model Testing -> Prediction]